

Ghost Identity Hunter — Tooling Inventory, Installation Guide and Provider Gap

Author: Dhayanidhi Rajasekaran **Repository:** <https://github.com/dhayarajas/gih> **Date:** 2026-08-02

1. Scope

Part A lists every tool and service Ghost Identity Hunter (gih) actually invokes, with the installation command for each and the check that proves it is wired up. Part B lists the Maltego data providers gih does **not** use, so the coverage boundary is explicit.

Sources: docs/T00L_COVERAGE.md, docs/EXTERNAL_T00LS.md, config/config.yaml, requirements*.txt, Dockerfile.kali, and the published Maltego Transform Hub catalogue (<https://www.maltego.com/transform-hub/>).

2. Base installation

Python 3.10+ is required.

```
git clone https://github.com/dhayarajas/gih.git
cd gih
pip install -r requirements.txt
```

Runtime Python dependencies (all installed by the command above): click, networkx, phonenumbers, requests, Brotli, beautifulsoup4, Pillow, pyvis, Jinja2, tabulate, pyyaml, numpy, urllib3.

Optional extras, each degrading cleanly when absent:

```
pip install -r requirements-dev.txt      # pytest, pytest-cov, ruff
pip install -r requirements-optional.txt # face matching (dlib, ~3 GB) + Neo4j driver
```

Verify the external tool inventory at any time — this is the authoritative check and prints exactly what the host has:

```
python -m src.cli investigate --check-tools
```

Part A — Tools gih uses

3. Integrated external tools (14)

Each of these has a real output parser in src/modules/external_tools.py and is dispatched by _process_external_tools. A missing binary is never scheduled; the investigation continues without it.

#	Tool	Used for	Install command	Verify
1	sherlock	username → account presence	pip install sherlock-project	sherlock --version
2	maigret	username → account presence (wider site list)	pip install maigret	maigret --version
3	osrframework (usufy)	username → social account presence	pip install osrframework	usufy --help
4	holehe	email → registered services	pip install holehe	holehe --help
5	theHarvester	domain → emails +	Kali: sudo apt install theharvester · elsewhere: pipx install git+https://github.com/laramies/theHarvester	theHarvester -h

#	Tool	Used for	Install command	Verify
		subdomains		
6	subfinder	domain → subdomains	go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest	subfinder -version
7	sublist3r	domain → subdomains	pip install sublist3r (Kali also packages it as sublist3r)	sublist3r -h
8	amass	domain → subdomains (passive)	Kali: sudo apt install amass · elsewhere: go install -v github.com/owasp-amass/amass/v4/...@master	amass -version
9	whois	domain → registrar, dates, registrant	sudo apt install whois	whois example.com
10	whatweb	domain → web technologies, resolved IP	sudo apt install whatweb	whatweb --version
11	nmap	IP → open ports and service versions	sudo apt install nmap	nmap --version
12	shodan	IP → host intelligence (<i>API key required</i>)	pip install shodan then shodan init <API_KEY>	shodan info
13	exiftool	image → GPS, camera, timestamps	sudo apt install libimage-exiftool-perl	exiftool -ver
14	wayback_machine	domain → archived URLs	none — HTTP call to web.archive.org CDX API	reachable network

Notes worth knowing before relying on a row:

- osrframework exposes no binary of its own; gih detects and runs usufy. It is restricted to the social platform tag with 32 threads (full list far exceeds the time budget; its timeout is 180 s).
- amass -passive often exceeds its 60 s budget and returns nothing; subfinder, sublist3r and theHarvester cover the same role.
- wayback_machine needs no binary and reports available whenever the network is up, but its CDX query commonly takes 9–31 s against a 30 s timeout, so archived URLs are absent from many runs.
- nmap runs -Pn -F -sV --version-light, i.e. no -sS, so **no root privileges are needed**.
- shodan reads the key stored by shodan init; no API key is ever placed on a command line.

4. Built-in modules and plugins (no installation)

These ship with the repository and need only the Python dependencies above.

Capability	Module	Notes
Phone validation / carrier / region	src/modules/phone_osint.py	uses phonenumbers
Email OSINT and breach lookup	src/modules/email_osint.py, email_breach_plugin	Have I Been Pwned; key optional
Username enumeration (built-in)	src/modules/username_search.py	independent of sherlock/maigret
Full-name seed expansion	orchestrator + investigation.strict_match	generates username candidates
Google Dorks	src/modules/google_dorks.py	Custom Search API when keyed, scraping otherwise
Image EXIF, hashing, reverse-search links	src/modules/image_search.py	link generation only, no image API
Face similarity	src/modules/image_match.py	needs requirements-optional.txt (dlib)

Capability	Module	Notes
LeakOSINT breach records	src/modules/leakosint.py	paid token required; leads the report when present
Correlation, confidence, risk scoring	src/correlation/linker.py	core, always on
Interactive graph	src/graph/visualizer.py	pyvis / networkx
HTML / JSON / CSV / PDF reports	src/reporting/	4 HTML templates + redaction
Optional graph backend	Neo4j	pip install neo4j, then use_neo4j: true

5. API keys

No key is ever written to a command line, a report, a log, or the database.

Service	Configuration	Behaviour without it
Shodan	plugins.shodan.api_key or SHODAN_API_KEY, plus shodan init	plugin reports unavailable, run continues
LeakOSINT	plugins.leakosint.api_key, or LEAKOSINT_API_TOKEN / LEAKOSINT_API_KEY	skipped, no Breach Records section rendered
Have I Been Pwned	plugins.email_breach.api_key	optional; unauthenticated calls are rate-limited
Google Custom Search	google_api_key + google_cx	falls back to scraping
Etherscan, GeoNames	plugins.*.api_key	declared only — no integration exists , never scheduled

6. Declared but deliberately not invoked (21)

ToolChecker knows 36 tools; 14 are integrated and 1 (leakosint) runs through the plugin path. The rest are detected and reported by `--check-tools` but never dispatched, each with a reason recorded in `UNIMPLEMENTED_TOOLS`:

social_analyzer, emailharvester, recon-ng, spiderfoot, ghunt, photon, metagoofil, etherscan, geonames, wappalyzer, masscan, nikto, sqlmap, curl, wget, nslookup, dig (DNS records describe the mail/web provider, not the seed), tor_browser, flagfox, user_agent_switcher, google_dorks (handled in-process, not as a subprocess).

7. One-command environment (recommended)

The Kali image installs the whole tool chain, so nothing above has to be installed by hand:

```
docker build -f Dockerfile.kali -t gih:kali .
docker run --rm -it -v "$HOME/.ghost_hunter:/home/ghosthunter/.ghost_hunter" gih:kali
```

Equivalent bare-metal one-liner for a Debian/Ubuntu host:

```
# in the Ubuntu/Debian repositories
sudo apt update && sudo apt install -y nmap whois whatweb libimage-exiftool-perl python3-pip golang-go

# PyPI
pip install sherlock-project maigret holehe sublist3r osrframework shodan

# not packaged for Ubuntu – install from source
pipx install git+https://github.com/laramies/theHarvester
go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest
go install -v github.com/owasp-amass/amass/v4/...@master
```

On Ubuntu 22.04 apt carries nmap, whois, whatweb and libimage-exiftool-perl but **not** amass, theharvester or sublist3r — those three are packaged only on Kali, which is why the Docker image is the shorter route.

Part B — Maltego providers gih does not use

Of the 99 providers published on Maltego's Transform Hub, gih reaches four (**Have I Been Pwned**, **Shodan**, **Wayback Machine**, **Dorking**) and has weaker or stub equivalents for three (**Google Social Network Transforms** → sherlock/maigret;

TinEye / Image Analyzer → reverse-search links only; **Etherscan** → declared, no plugin). The remaining 92 have no path in gih today.

Malware / sandbox / threat intelligence

- VirusTotal Public API
- VirusTotal Premium API
- Hybrid Analysis
- Intezer Analyze
- PolySwarm
- Cisco Threat Grid
- CrowdStrike ThreatGraph
- CrowdStrike Intel
- Mandiant
- Abuse.ch URLhaus
- ThreatCrowd
- ThreatMiner
- ThreatConnect
- OpenCTI
- ATT&CK-MISP
- Anomali
- Intel 471
- Recorded Future
- Cybersixgill

Vulnerabilities

- NIST NVD
- Vulners
- Censys
- GreyNoise Enterprise
- GreyNoise Community
- Elemendar
- Team Cymru

Infrastructure, DNS and passive data

- DomainTools Iris Investigate
- DomainTools Enterprise
- Farsight DNSDB
- ZETalytics Massive Passive
- WhoisXML API
- Host.io
- IPInfo
- MaxMind
- PeeringDB
- SSL Certificate Transparency Transforms
- DNSTwist
- alphaMountain
- HYAS Insight
- CrowdSec
- Criminal IP
- Spamhaus Intelligence
- urlscan.io
- SOCRadar
- Silobreaker

Breaches and leaks

- SpyCloud Cybercrime Investigations
- Constella Intelligence
- Darkside
- Hades
- Flashpoint
- Digital Shadows

- ZeroFOX
- Cofense Intelligence

Deep and dark web

- DarkOwl
- Vysion
- Web-IQ
- Corvus Intelligence
- Crowlingo
- Tisane Labs

People and personal identifiers

- Pipl
- FullContact
- PeopleMon
- PhoneSearch
- Epieos
- OSINT Industries
- Hunter.io
- Clearbit
- LittleSis
- LoginsoftOSINT
- FlightAware
- Espy
- Social Links Professional
- Social Links CE
- Google Maps Geocoding

Company, corporate and sanctions data

- OpenCorporates
- Orbis – Bureau Van Dijk
- OCCRP Aleph
- OpenSanctions
- EntitiesUA

Cryptocurrency

- Tatum Blockchain Explorer
- Crystal Intelligence

Phishing and reputation

- Scamadviser
- AbuseIPDB
- AlienVault OTX

Image, content and language

- GeoSpy
- DeepL
- Regex Library

Interoperability, case management, long tail

- STIX 2 Utilities
- Polonious
- Discogs
- The Movie Database

Cheapest of these to add

Free or free-tier, so closable without a commercial contract: **crt.sh / Certificate Transparency**, **IPInfo** (or MaxMind GeoLite2), **NIST NVD**, **VirusTotal Public API**, **Hunter.io**, **AbuseIPDB**, **Abuse.ch URLhaus**, **urlscan.io**, **OpenSanctions**, **Etherscan** (already declared), **STIX 2 / MISP export**.

Two model changes unlock most of them: `file_hash` and `cve` artifact types, and an organisation entity. Without those, several of the free providers have nowhere to attach their results.

Structurally unreachable without commercial contracts: SpyCloud, Constella, DomainTools, Farsight, Recorded Future, Flashpoint, Cybersixgill, DarkOwl, Intel 471, Mandiant, CrowdStrike, Pipl, Social Links, Orbis, HYAS, Silobreaker, Digital Shadows, ZeroFOX.